

An adaptive training-less framework for anomaly detection in crowd scenes



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ABSTRACT

Anomaly detection in crowd videos has become a popular area of research for the computer vision community. Several existing methods have determined anomaly as a deviation from scene normalcy learned via separate training with/without labeled information. However, owing to rare and sparse nature of anomalous events, any such learning can be misleading as there exist no hardcore segregation between anomalous and non-anomalous events. To address such challenge, we propose an adaptive training-less system capable of detecting anomaly on-the-fly. Our solution pipeline consists of three major components, namely, adaptive 3D-DCT model for multi-object detection-based association, local motion descriptor generation through an improved saliency guided optical flow, and anomaly detection based on Earth mover's distance (EMD). The proposed model, despite being training-free, is found to achieve comparable performance with several state-of-the-art methods on publicly available UCSD, UMN, CUHK-Avenue and ShanghaiTech datasets.

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1. Introduction

Anomaly detection in crowd videos has evolved as an important surveillance problem among the computer vision community. However, detection and localization of anomaly is a challenging task as the notion of anomaly is subjective and context-dependent [1,2]. For instance, person running in sports court is normal while the same act may be abnormal outside a bank. In general, an event can be defined as anomalous if its occurrence is rare or unexpected [1]. Acquisition of abnormal events are much difficult and sometimes infeasible compared to normal ones owing to rarity and sparsity of occurrence within a video. This inadequacy of sufficient representative set for abnormal events limits the application and effectiveness of traditional supervised learning approaches in solving the anomaly detection problem. Most methods [3–7] on crowd anomaly detection usually learn a similarity model based on ordinary (normal) events and label a test event anomalous which significantly deviates from the normal ones. Such models are learned from a) determining characteristic motion and appearance and b) modeling changes in such motion and appearances. Traditionally, anomaly detection based on spatio-temporal frequency [6,7], mixture of dynamic textures (MDTs) [5] and spatio-temporal motion context [8] are found to be effi-

cient. Other methods extract characteristic motion features from different optical flow based methods, such as, social force models [9], trajectory based techniques [3], histograms of optical flow (OF) [9,4], histogram of oriented gradients [10,11], and spatial saliency of the motion feature [4]. Although, these methods have shown their efficacy in detecting anomaly, more abstract semantic feature mining and utilization are clearly lacking. Recently, with advances in deep learning techniques, researchers have explored their applicability in anomaly detection problems. Methods like [12,13] developed deep Gaussian mixture model for learning normal events while other methods like [14,15] applied adversarial learning to discriminate normal frames from the abnormal ones. Some approaches also developed joint models of deep and traditional approaches. For instance, Luo et al. [2] combined sparse coding with Recurrent Neural Network (RNN) framework while Xu et al. [16] applied deep learning over video patches for anomaly detection.

Several methods were also proposed (both classical [17,18] and deep-learning [19,20] based) in past few years that can perform in real-time. However, these methods suffer from certain drawbacks despite their real-time performance. While the deep learning methods [19,20] have gained state-of-the-art performance with high computational costs (i.e., requiring several-core processors and high-end GPUs), the classical methods ([17,18]) can run with low computational resources but with inferior results. In sharp contrast to both these types of approaches, our framework better

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balances the accuracy vs. computational resource trade-off. So, we have achieved satisfactory results with minimal computational resources. Moreover, our model performance can further be boosted with GPU acceleration. Secondly, training a supervised model is always difficult due to heavy class imbalance [21] as number of abnormal frames are found to be much less compared to the normal ones. In contrast, a semi-supervised technique constructs a normalcy (normal patterns) model [22] from a given normal training dataset. Although the approaches of one-class semi-supervised paradigm alleviates the data scarcity issue of supervised models in some sense yet they possess some fundamental disadvantages of their own. The disadvantages are the following – a) the hierarchical features extracted within deep hidden layers may not well represent fewer anomalous instances and may lead to overfitting [23], and b) Lu [24] theoretically established that there cannot be an optimal semi-supervised algorithm just by knowing the distribution of unlabeled data without making any assumption between the labeled and unlabeled data distribution. Our method being training-less is free from such biases and assumptions and thereby can perform well in diverse crowd scenes.

In contrast to above learning based methods, certain methods reported in literature [11,25,26] can perform anomaly detection without requiring explicit training. Such methods usually adapt/fine-tune their model parameters while deriving inference about a video sequence. This category of methods can be termed as “training-less” models. These models have certain advantages over training-based models, such as, a) these models can be potentially applied to a scene without any prior notion of scene normalcy, b) with proper adjustments such models can invariably handle multiple unseen abnormality without any bias, and c) constant update of model parameters leads to more automated scene adjustments with less human assistance.

In this paper, we propose an adaptive training-less system for detecting anomaly in crowd videos. Please see Fig. 1. Firstly, several object proposals are generated using our pedestrian detection model [27]. Proposals in form of bounding boxes (BBs) are used to extract 2D templates of objects. These object proposals are fed into a pre-trained Region-based Convolutional Neural Network (RCNN) [28] for semantic feature extraction in conjunction with object proposal generation. Proposals of same objects are associated over frames based on appearance similarity using our proposed 3D-DCT approach and extracted RCNN feature. A 3D-stack of templates is preserved for each object over the frames of a scene. Several local motion descriptors are generated over these objects. The changes between these descriptors are computed using Earth mover's distance (EMD) [29] over consecutive frames. An aggregate change of all the descriptors exceeding a certain threshold mark the presence of the anomaly. Our main contributions are now summarized below:

1. We have developed an adaptive 3D-DCT model based on a self-adjustable scaling parameter for multi-pedestrian detection and association. Furthermore, we have integrated semantic information mined from pre-trained RCNN to enhance the robustness of association.
2. We have also proposed saliency modulated optical flow (OF) map generation via a reward-penalty function to generate more discriminatory local motion descriptors. Saliency, by itself, is difficult to associate with anomaly, but it can add a significant cue to OF map for yielding more discriminatory motion descriptor which can help in detecting anomaly.
3. Overall, we present an adaptive training-less system that can perform crowd anomaly detection without any prior knowledge. We later demonstrate in the paper that state-of-the-art performance has been achieved on several publicly available datasets.

The rest of the paper is organized in the following manner: in Section 2, we discuss the related works. In Section 3, we describe our proposed method. We present the experimental results in Section 3. Finally, the paper is concluded in Section 5 with an outline for directions of future research.

2. Related work

Existing works on crowd anomaly detection [1,46,2,47] can be grouped into several categories [48], namely, a) fully-supervised [15,12], b) semi-supervised [39,38], c) unsupervised [10,26], and d) training-less approaches [25,7]. A summary of these four different types of methods is described in Table 1. While former three approaches require training (with or without utilizing label) for modeling crowd behavior, training-less approaches apply external (*a priori*) domain knowledge to create rules/models for separating normal and abnormal classes.

2.1. Fully-supervised approaches

Methods in this category require both “normal” and “abnormal” labeled data set to train a model. For instance, Cui et al. [30] detected and tracked several key points based on Interaction Energy Potentials (IEPs) and learned a Support vector machine (SVM) to separate trajectories of normal from abnormal ones to detect anomaly. Cong et al. [8] constructed Edge orientation histogram (EOH) and Multi-layer histogram of OF (MHOF) as appearance and motion descriptor respectively, whereas Mahadevan et al. [5] extracted spatio-temporal patches to learn mixture model of dynamic textures (MDTs). Ullah et al. detected and localize anomalous entries by integrating spatial and temporal descriptor into

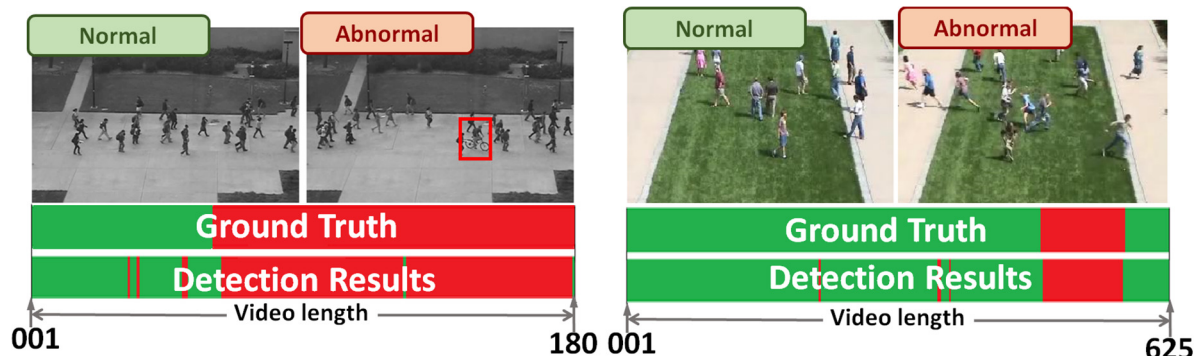


Fig. 1. Anomaly detection in UCSD and UMN datasets using the proposed method. The length of the color bar indicates the duration of the videos where green and red regions indicate non-anomalous and anomalous regions respectively.

Table 1

Categorization of State-of-the-art anomaly detection methods. The “Type” column shows the type of the anomaly that the various methods detected.

Category	Reference	Feature	Model	Type	Dataset
Fully-Supervised	Mahadevan et al. [5]	Dynamic textures	Mixture models	NP	UCSD
	Cui et al. [30]	Spatio-temporal key points	SVM	EP, SA	UMN, BEHAVE
	Cong et al. [8]	Spatio-temporal Motion	Mixture models	NP	UCSD
	Biswas and Gupta [31]	CNN	Sparse component tracking	NP, EP	UCSD, UMN
	Luo et al. [2]	CNN	stacked RNN	NP, SA	UCSD ped2, Avenue
	Ravanbakhsh et al. [14]	Motion	GANs	NP, EP	UCSD, UMN
	Luo et al. [32]	CNN	ConvLSTM + AE	NP, SA	UCSD, Avenue, Subway
	Sabokrou et al. [15]	CNN	Sparse AE	NP, SA	UCSD Ped2, Subway
	Ullah et al. [17]	Spatio-temporal	R-CRF	NP, EP, FD	UCSD, UMN, UCD
	Ionescu et al. [19]	Appearance and motion gradient	object-centric AE	NP, EP, SA	UCSD Ped2, Avenue, UMN, ShanghaiTech
	Lin et al. [33]	Spatio-temporal	SF + Attention Module	Abuse, assault	UCF-Crime
	Khan et al. [34]	Histogram of superpixel	Gaussian discriminant analysis	NP, EP, TA	UCSD, UMN, LV
	Zhou et al. [35]	Fused Motion	AnomalyNet	NP, EP, SA	UCSD, UMN, Avenue
	Singh et al. [36]	CNN	Ensemble of CNNs	NP, SA	UCSD, Avenue
Semi-Supervised	Bertini et al. [37]	Spatio-temporal	k-means trees	NP	UCSD
	Hasan et al. [38]	CNN and motion	Convolutional AE	NP, SA	UCSD, Avenue
	Liu et al. [39]	Intensity and Motion	U-net based GAN	NP, SA	UCSD, Avenue, ShanghaiTech
	Hao et al. [40]	Spatio-temporal	Texture classification	EP	UMN
	Hatirnaz et al. [41]	Motion	Concept based semantic search	EP, FD	UMN, PETS2009
Unsupervised	Kim and Grauman [42]	Motion	MRF	SA	Subway
	Xu et al. [16]	Motion	AE + SVM	NP, SA	UCSD, Train
	Sabokrou et al. [43]	Spatio-temporal	Sparse AE	NP, EP	UCSD ped2, UMN
	Li et al. [44]	Spatio-temporal	Dictionary Learning	NP, EP	UCSD, UMN
	Ionescu et al. [26]	Spatio-temporal and CNN	SVM classifier	NP, EP, SA	UCSD, UMN
Training-less	Hu et al. [10]	Motion	Dictionary Learning	NP, SA, FD	Avenue, Subway
	Bansod et al. [18]	HoMM	K-means Clustering	NP, EP	UCSD, UMN, Avenue, PETS2009
	Mehran et al. [9]	Motion	SF	EP	UCSD, UMN
	Zhao et al. [11]	Motion	Dynamic Sparse Coding	SA	UMN
	Roshthkhari et al. [7]	Spatio-temporal	BoVW	NP	Subway
	Yuan et al. [45]	Structural Motion	3D-DCT	NP, EP	UCSD
	Giorno et al. [25]	Aggregated Motion	SVM classifier	SA	UCSD, UMN
	Nawaratne et al. [20]	CNN	Active Learning with Fuzzy Aggregation	NP, SA	Avenue
					UCSD, Avenue

Recurrent Conditional random field model. Khan et al. utilized histogram of superpixel (HoS) to learn Gaussian discriminant analysis (GDA) model to cluster anomalous frames. Several Models [4,49,31,2] based on spare representation is also proposed for anomaly detection. For instance, Biswas and Gupta [31] tracked sparse components via Kalman filtering while Cong *et al.* [4] construct an over-complete spare dictionary to separate normal from abnormal ones. In a recent work Luo *et al.* [2] encoded consecutive frames with similar nature via Temporally-coherent Sparse Coding (TSC). These encoded frames are then fed to a stacked RNN to facilitate anomaly detection. Recently, several deep learning models [15,12,14] have evolved for detecting anomaly. Sabokrou *et al.* [15] constructed a fully connected network for fast anomaly detection whereas Ravanbakhsh *et al.* [14] introduced Generative Adversarial Networks (GANs) to discriminate frames with abnormal motion patterns by comparing them with their generated normal counterpart. Ionescu *et al.* [19] employed a two stage training to learn an object-centric autoencoders with an SVM classifier. While the autoencoders learn to detect objects, SVM is used to separate abnormal from normal ones. Luo *et al.* [32] utilized autoencoders to integrate CNN based appearance features with Convolutional Long Short Term Memory (ConvLSTM) to predict future anomaly. Lin *et al.* performed a decision level fusion of appearance and motion within each frame via Attention and social force (SF) model respectively whereas Singh *et al.* achieved anomaly detection via learning Ensemble of several CNN models. In general, supervised

models generally work well if the class distributions or the class boundaries are well-known to the model. Data scarcity and class imbalance [21] is a major challenge in learning such model as the abnormal set is much smaller compared to the normal one. Also, most supervised methods cannot be generalized in varying contexts of different anomalous scenes as they require explicit labeling and training for each scene.

2.2. Semi-supervised approaches

Methods in this category require less labeled set [22] to train a model as compared to the supervised ones. In this approach usually a normalcy model is learned only from a given normal training set. This approach of one-class semi-supervised learning paradigm [21] can also be termed as weakly supervised model. For instance, Zhang *et al.* [50] utilized Markov random field (MRF) to model normal events and adapted a Bayesian framework to separate abnormal events. Bertini *et al.* [37] utilized k-means trees to progressively detect anomaly in test videos initialized with a small portion of normal video sequence. Hasan *et al.* [38] has adopted hand-crafted motion features with deep features to learn temporal regularity of a video using Convolutional autoencoder. A generated frame having high reconstruction error with the actual ones is deemed anomalous. Liu *et al.* [39] proposed an anomaly prediction framework using U-net based GAN. Recently, Hatirnaz *et al.* [41] developed concept based semantic search for detection of anom-

ally. Generative methods [39] in this category also follow inductive learning where unlabeled samples are labeled based on a small labeled set. Although, such approaches can handle data scarcity issues due to its one-class (normal) learning paradigm [22], they have an inherent disadvantage stemming from the assumed relationship [24] between the labeled and unlabeled data distribution for model learning.

2.3. Unsupervised approaches

Approaches in this category detect anomaly based on an underlying assumption that majority of data in an unlabeled set follow similar trend and the data that fits least is deemed anomalous. Models learned based on this approach separate data by exploring the inter-data relationships. For instance, Kim and Grauman [42] learned local OF patterns based on mixture of probabilistic PCA (MPPCA) and represented such patterns using space–time markov random field (MRF) graph. Sabokrou et al. [43] described spatio-temporal video cubes using local and global descriptors and adopted a Sparse autoencoder to determine the structure similarity of adjacent patches. Li et al. [44] also utilized spatio-temporal video cubes to perform unsupervised dictionary learning to detect anomaly. Ionescu et al. [26] adopted an iterative training for binary classification of sliding window which is then used to unmask the abnormal segments from the normal ones. Recently, novel histogram of contextual gradients (HOCG) have been proposed by Hu et al. [10] to better characterize the composition of anomalous events. Bansod et al. [18] on the other hand, developed Histogram of magnitude and moments (HoMM) to cluster normal and abnormal frames. Methods in this category curtail some drawbacks of supervised and semi-supervised approaches as no label is required for model learning. However, they suffer from a strict assumption that all anomalous events are distinct from the normal ones. So, such models can fail if the distribution of normal and abnormal frames are not well separated.

2.4. Training-less approaches

Methods under this category do not require training data either in the labeled or unlabeled form. Rather, these models are based on external domain knowledge [48]. Learning in such model is achieved on-the-fly based on parameter adjustments/adaptation during inferences on test video. For instance, Mehran et al. [9] developed social force model over OF using particle advection whereas Zhao et al. [11] performed real-time unusual event detection via online dictionary learning through dynamic sparse coding. Roshtkhari et al. [7] constructed bag-of-video-words (BOV) from densely constructed spatio-temporal volumes (STVs) and determined low likelihood of occurrence via online Fuzzy clustering. Nawaratne et al. [20] performed active learning with Fuzzy Aggregation of CNN features to achieve real-time performance. Yuan et al. [45] proposed an particle interaction based structural context descriptor (SCD) to represent crowd anomaly. Lately, Giorno et al. [25] learned low probability events via classical density estimation without requiring temporal ordering of video frames but the need to permute test video frames before making decisions thereby restricting them to process online. Though these approaches curb out the issue related to model learned from labeled/unlabeled data, their success heavily depend upon the appropriateness about the external knowledge of a given target scene [48]. One major limitation of these methods are that they do not well adapt to the dynamics of the scene. This insensitivity to local or minor changes of a scene make them typically static [48] with unsatisfactory performance. Moreover, these methods cannot be readily generalized/applied to variety of scenes/dataset (as evident in Table 1) whereas our modeling is robust and applicable to diverse scenes.

In summary, the performance of current supervised and semi-supervised anomaly detection methods are limited by issues related to data scarcity and/or pre-assumed relationship between labeled and unlabeled data while unsupervised models rely mostly on the distinctive nature of normal and abnormal sets based on distribution. Available training-less methods on the other hand are free from such biases and assumptions but they rely heavily on fixed domain knowledge and are thereby unable to adjust with the dynamics of the scene. In sharp contrast to aforementioned methods, our training-less model is robust in adapting well with the dynamics of the scenes and is free from any biases and assumptions related to data/scene. Thus, our approach is able to achieve state-of-the-art performance.

3. Proposed framework

In this section, we provide a detailed description of our proposed method. The three major components of our solution are: A) Multi-object association-based detection, B) Local structural descriptor generation, and C) Crowd anomaly analysis. A schematic diagram of the overall pipeline is shown in Fig. 2.

3.1. Multi-object association-based detection

Multi-object detection-based association is composed of three major steps, namely, a) Object proposals with RCNN feature extraction, b) Adaptive 3D-DCT based association, and c) Object preservation via template pool building. These steps are discussed below in details.

3.1.1. Object proposals with RCNN feature extraction

We have extracted RCNN based semantic features for better associating objects under motion but in a training-less fashion. In a RCNN network, a ROI (Region-of-interest) pooling is performed first with the help of Region-Proposal-Function (RPF) [28]. Then, the pooled area is passed through pre-trained Resnet50 [51] followed by two fully connected layers for feature extraction. Generally, RPF is learned during RCNN training. To achieve this task of feature extraction without training we replace RPF by our training-less pedestrian detection [27] approach for generating object proposals. In this approach, an edge-guided watershed segmentation strategy is performed over the foreground mask generated by Gaussian mixture model (GMM). Although k-means based clustering [52] is simple, GMM is widely used in different areas [53–55], due to two potential advantages. Firstly, GMM takes variance of data distribution into account and therefore perform better clustering for any shape of distribution (of the data). Secondly, k-means suffers from initialization problem, i.e., the quality of clustering is highly dependent on the choice of initial cluster centers (or seed points). We applied GMM due to its better performance in the proposed method. Several proposal BBs of varying sizes are generated over these segmented regions. Out of these BBs, valid candidates are selected based on two thresholds, namely, area (ϕ_{thres}^{BB}) and entropy (ψ_{thres}^{BB}) [27] as final proposals for further processing.

3.1.2. Adaptive 3D-DCT based association

We design an alternative version of 3D-DCT tracker [56] for associating multiple objects in an unsupervised fashion. The model preserves a 3D-volume of object images (templates) into a stack (template pool) from the past frames in a video. Separate template pools are built incrementally for every object in a scene. Each template pool is updated iteratively with a restricted stack size to preserve the best appearance of object for obtaining better associates from future frames. Suppose there are N proposal BBs in current

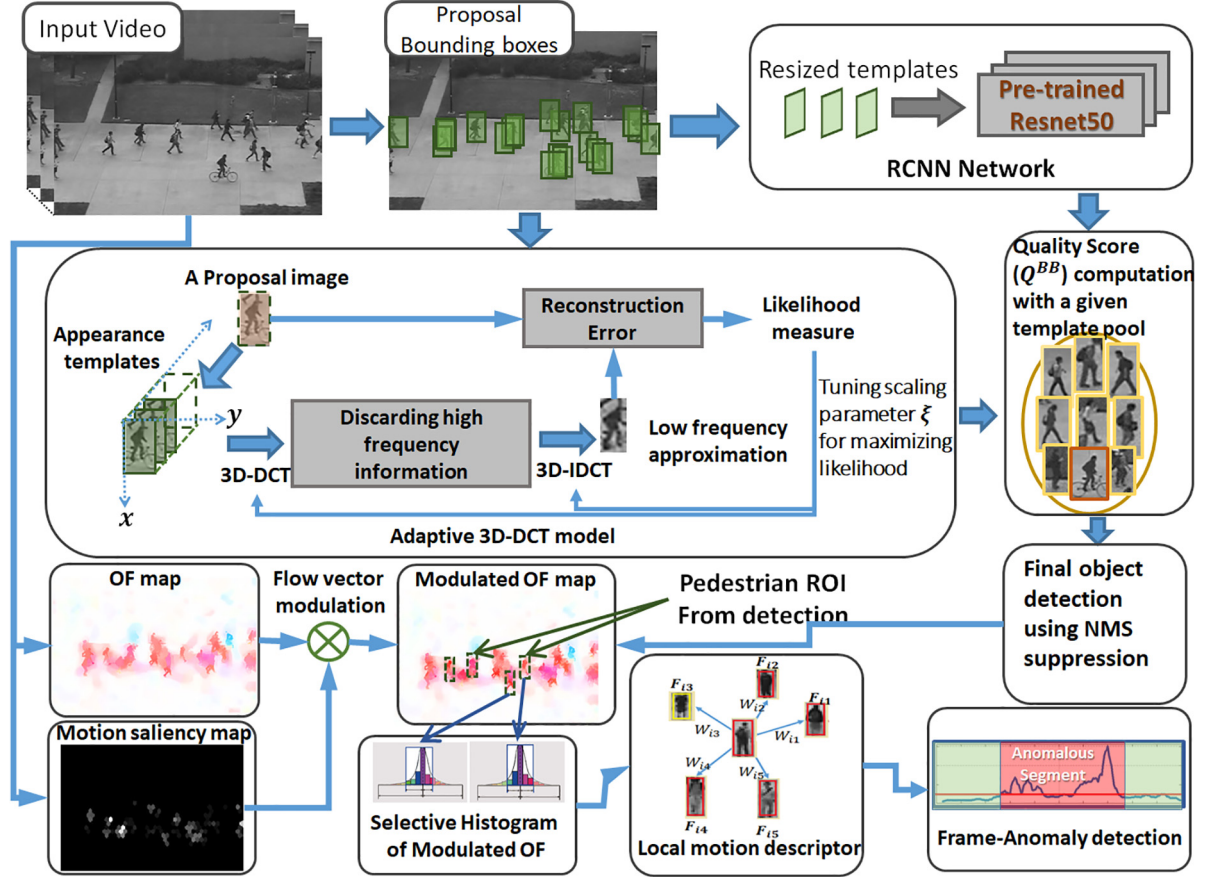


Fig. 2. An illustrative overview of our proposed training-less framework for anomaly detection via adaptive 3D-DCT based model and motion descriptor generated over selective histogram of modulated Optical Flow (SHMOF).

frame with p template pools built over past frames in ΔT span of time. Then, each $\kappa \in p$ template pool can be represented as $\{S_{\text{III},\kappa}(x,y,t)\} \in \mathbb{R}^{W \times H \times N_{\text{temp}}}$ where, (x,y,t) are the coordinates of the 3D volume with W, H, N_{temp} respectively denoting the width, height and stack size of a pool. The corresponding 3D-DCT coefficient of $S_{\text{III},\kappa}$ is denoted by $C_{\text{III},\kappa}$ and is computed as in [56]. A low frequency approximation of $S_{\text{III},\kappa}$ (denoted by $S_{\text{III},\kappa}^*$) can be computed by performing an inverse 3D-DCT operation on $C_{\text{III},\kappa}$ (obtained by setting the coefficients of high frequency components of $C_{\text{III},\kappa}$ to 0). The coefficients corresponding to higher frequencies are usually sparse and mostly consist of texture cues while the low frequency coefficients are relatively dense and contribute mostly to the appearance of the object. Thus, for a new template $n_{\kappa} \in N_{\kappa} \subset N$ (set N_{κ} has non-zero spatial overlap with the last template of $S_{\text{III},\kappa}$, i.e., $S_{\text{III},\kappa}(:, :, N_{\text{temp}})$) the reconstruction error with $S_{\text{III},\kappa}$ can be computed as $\|n_{\kappa} - S_{\text{III},\kappa}^*(:, :, N_{\text{temp}} + 1)\|^2$. A likelihood measure of n_{κ} with template pool $S_{\text{III},\kappa}$ can be obtained based on the following equation:

$$\mathcal{L}_{\kappa}(n_{\kappa}, \xi) = \exp\left(-\frac{1}{2\xi} \|n_{\kappa} - S_{\text{III},\kappa}^*(:, :, N_{\text{temp}} + 1)\|^2\right) \quad (1)$$

where, ξ is the scaling factor. The likelihood function in Eq. 1 is normalized over the sigmoid function $\rho(x) = 1/(1 + e^{-x})$, $\rho(x) \in [0, 1]$. The scaling parameter ξ is adaptively optimized over dynamically selected set N_{κ} to make the likelihood of each $n_{\kappa} \in N_{\kappa}$ with $S_{\text{III},\kappa}$ more discriminative. An optimal ξ will maximize the spread of likelihood function $\mathcal{L}$ over the sigmoid function $\rho(x)$. An optimized ξ_{κ} can be computed based on the equation:

$$\xi_{\kappa} = \underset{\xi}{\operatorname{argmax}} \{ \operatorname{var}(\rho(\mathcal{L}_{\kappa}(n_{\kappa}, \xi))) \} \quad (2)$$

where, $\operatorname{var}(\cdot)$ is the variance. This dynamic adjustment of scaling parameter ξ is much more effective for discriminative score generation as arbitrarily smaller or larger ξ will concentrate normalized likelihood values towards 0.5 or 1 which in turn would lead to false associations. The mean optimal $\hat{\xi}$ is computed based on these optimal sets (ξ_{κ}) of each κ^{th} template pool in the following manner:

$$\hat{\xi} = \frac{1}{p} \sum_{\kappa=1}^p \xi_{\kappa} \quad (3)$$

To further enhance multi-object association, we have computed a quality score based on both likelihood and semantic features extracted from RCNN of a template $n_q \in \bigcup_{\kappa} N_{\kappa}$ and the last sample of the κ^{th} template pool. Thus, the semantic distance between n_q and $S_{\text{III},\kappa}(:, :, N_{\text{temp}})$ can be represented as:

$$D_{q,\kappa}^{\text{RCNN}} = \|f_{\text{RCNN}}(n_q) - f_{\text{RCNN}}(S_{\text{III},\kappa}(:, :, N_{\text{temp}}))\|_2 \quad (4)$$

where, $f_{\text{RCNN}}(\cdot)$ is the semantic deep feature extracted from RCNN. Thus, the q^{th} template is associated with the κ^{th} template pool if it attains a highest quality score $Q_{q,\kappa}^{\text{BB}}$ as:

$$Q_{q,\kappa}^{\text{BB}} = \max_{\kappa} \{ D_{q,\kappa}^{\text{RCNN}} * \rho(\mathcal{L}(n_q, \hat{\xi})) \} \quad (5)$$

Most of the generated proposals are redundant and overlapping. To keep the best matches between the competing overlapping templates, we apply non-maximal suppression (NMS) based on

quality score $Q_{q,\kappa}^{BB}$. The bounding boxes remaining after NMS are the final detections of the current frame. Since, detection is achieved after finding proper associates with the current template pools, we term this mechanism as association based detection scheme.

Algorithm 1: Multi-Object Association Based Detection

Input: Video Sequence

Output: β

Initialization: $\phi_{thres} \leftarrow 0.5, \psi_{thres} \leftarrow 0.7$

```

1: for  $t = 1$  to  $T$  /*  $T$  is the total No. of frames */
2:   Obtain foreground region using GMM 3:   Perform
   edge-fused region segmentation 4:   Generate proposal
   bounding boxes satisfying  $\phi^{BB} > \phi_{thres}$  and  $\psi^{BB} > \psi_{thres}$ 
5:   if  $t == 1$  then
6:     Apply NMS over  $N$  templates
7:     Initiate template pools based on Eq. 7
8:   else
9:     Compute  $Q^{BB}$  score based on Eq. 5
10:    Apply NMS over  $N$  templates
11:    Update template pools based on Eq. 6 and 7
12:   end if
13:   Obtain  $\beta$  observers
14: end for

```

3.1.3. Object preservation via template pool building

Let NMS in the frame t keeps $N' \subseteq N$ templates. Let β templates are associated with the previous template pools. Hence, remaining $(N' - \beta)$ templates form the newly generated template pools. If these β objects are associated with template pools that continue to exist over a duration ΔT , then they are deemed appearance-wise stable and are hence termed as observers. The rest of the detections are treated as targets. We finally discard template pools which remain unassociated over ΔT duration. Possible reasons behind no association include occlusion, exit, miss and false detection. Rejection of such template pools reduces the possibility of over-fitting (false associations). It further restricts the number of template pools which would have otherwise unnecessarily increased the computational overhead. Now, we discuss our strategy for template pool association.

The updation of a previously existing template pool $\mathcal{T}_{\kappa}^{t-1}$ by a new template n_{κ}^t at time t can be expressed as:

$$\mathcal{T}_{\kappa}^t = \left\{ \text{Concat} \left(\mathcal{T}_{\kappa}^{t-1}, n_{\kappa}^t \right) : S_{\text{III}}(x, y, t-1)_{N_1 \times N_2 \times N_3} \right. \\ \left. \longrightarrow S_{\text{III}}(x, y, t)_{N_1 \times N_2 \times (N_3+1)}, \kappa \in \beta \right\} \quad (6)$$

where, $\text{Concat}(\cdot)$ is a concatenation operation that adds a new template in template pool stack. For the remaining $(N' - \beta)$ templates a new template pool is created which can be expressed as:

$$\mathcal{T}_{\kappa}^t = \left\{ \mathcal{T}_{\kappa}^t : S_{\text{III}}(x, y, t)_{N_1 \times N_2 \times 1}, \kappa \in (N' - \beta) \right\} \quad (7)$$

So, the overall template pool set $\mathcal{T}^t$ in the current frame t can be represented as a union of two sets (which themselves are expressed as unions): $\left(\bigcup_{\kappa \in \beta} \mathcal{T}_{\kappa}^t \right) \cup \left(\bigcup_{\kappa \in (N' - \beta)} \mathcal{T}_{\kappa}^t \right)$. The first term denotes the number of template pools being updated (using Eq. 6) and the second term denotes the number of template pools being created (using Eq. 7). To reduce the computational burden related to future association, we further restrict the number of templates associated with any template pool to K . If the number of associated templates reaches $K + 1$, the most irrelevant one is discarded, based on the least likelihood measure given by:

$$n_{d,\kappa}^t = \underset{d \in [1, K+1]}{\text{argmin}} \exp \left(-\frac{1}{2\zeta} \|n_{d,\kappa}^t - S_{\text{III},\kappa}^*(:, :, d)\|^2 \right) \quad (8)$$

where, $n_{d,\kappa}^t$ is the d th sample which is to be discarded from the updated κ^{th} template pool and $S_{\text{III},\kappa}^*(:, :, d)$ is the low frequency approximation of the d th template computed from the inverse 3D DCT operations. In this way, we conform to reliability preservation of most likely appearances of an object. This step is necessary to dynamically adapt to the changing scene of the video and also to reduce unnecessary computational overhead and accumulated error due to increasing stack size of a template pool over time. The overall procedure of our multi-object detection with association step is described in Algorithm 1.

3.2. Local motion descriptor generation

We now present a temporal saliency guided optical flow map for accurately describing local motions in our video frames. The details of this formulation is presented below.

3.2.1. Generation of MOF

There exist several methods for determining temporal saliency [57,58]. We choose the recently proposed method of [58] because of its accuracy and execution speed. Let, the OF vector for each pixel (x, y) be denoted as $(\mathbf{u}(x, y), \mathbf{v}(x, y))$ [59] and the corresponding temporal saliency score be denoted as $s(x, y) \in [0, 1]$ [58]. We propose a temporal saliency modulated optical flow vector (MOF), denoted by $(\mathbf{u}'(x, y), \mathbf{v}'(x, y))$ using the modulating function $\mathcal{F}(s)$ as:

$$\begin{aligned} \mathbf{u}'(x, y) &= \mathcal{F}(s(x, y)) \cdot \mathbf{u}(x, y) \\ \mathbf{v}'(x, y) &= \mathcal{F}(s(x, y)) \cdot \mathbf{v}(x, y) \end{aligned} \quad (9)$$

The function $\mathcal{F}(s)$ modulates each OF component in $(\mathbf{u}, \mathbf{v})$ based on a parameter $\tau \in [0, 1]$. For any pixel, if the corresponding saliency value is greater than τ , the flow value is enhanced and if it is less than τ , the flow value is diminished (see Fig. 3 in this connection). This is mathematically represented as:

$$\mathcal{F}(s) = \begin{cases} e^{\frac{s-\tau}{2\tau}} & \text{for } s \leq \tau \\ e^{\frac{\tau-s}{2}} & \text{for } s > \tau \end{cases} \quad (10)$$

As the modulating function $\mathcal{F}$ enhances and diminishes the motion magnitude based on the saliency score, we term it as reward-penalty function. The parameter τ is determined adaptively at each frame based on area under the cumulative distribution function (CDF) for a random variable τ such that $\tau = \underset{\tau'}{\text{argmax}} [P(\tau' \leq s) \leq \alpha]$ for a threshold α , with $\tau' \in [0, 1]$. The nature of the function $\mathcal{F}$ for different values of τ is shown in Fig. 4. The range of modulation function (Eq. 10) is restricted to $[e^{-\frac{1}{2}}, e^{\frac{1}{2}}]$. This range is empirically determined based on the fact that a bigger range would unduly emphasize the enhanced flow values over the diminished ones while a smaller range will result in negligible modulation effects.

3.2.2. Generation of SHMOF

Let the histogram of MOF (HMOF) maps generated over i th and j th pedestrian region-of-interests (ROI) (based on detected BB as described in Fig. 3) be represented as H_i and H_j respectively. The histograms H_i and H_j are circularly shifted such that the bins with highest magnitude is aligned at center of the modified histograms $H_{i,\text{shifted}}$ and $H_{j,\text{shifted}}$ respectively. A selective HMOF (SHMOF) is generated by limiting the range of HMOF by a parameter $\zeta \in [0, 1]$ [45]. The setting $\zeta = 0$ selects only the bin with maximum magnitude (i.e., bin at center in case of $H_{i,\text{shifted}}$ and $H_{j,\text{shifted}}$) while $\zeta = 1$ selects

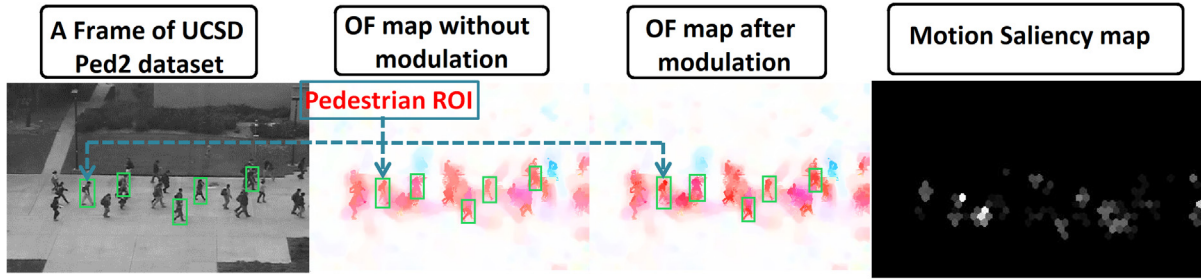


Fig. 3. An example of modulation in OF map represented in Middlebury color coding with (a) a frame/instance on ped2 dataset, (b) OF without modulation, (c) OF with modulation, (d) temporal saliency map. The figure illustrates the region belonging to pedestrian within bounding boxes (green boxes) that have undergone enhancement over suppression based on the motion saliency.

whole range of HMOF. We have set the value of ζ according to [45] in our experiments. The corresponding SHMOF representation for $H_{i,shifted}$ and $H_{j,shifted}$ are denoted by $H_{i,shifted}^\zeta$ and $H_{j,shifted}^\zeta$ respectively. Finally, a bin-to-bin histogram difference (Δh_{ij}^ζ) is computed between $H_{i,shifted}^\zeta$ and $H_{j,shifted}^\zeta$ using the χ^2 -distance which is expressed as:

$$\Delta h_{ij}^\zeta = \chi^2(H_{i,shifted}^\zeta, H_{j,shifted}^\zeta) \quad (11)$$

where, $\chi^2(H_i, H_j) = \frac{1}{2} \sum_{n=1}^{N_{bins}} \frac{(H_{i,n} - H_{j,n})^2}{H_{i,n} + H_{j,n}}$ and N_{bins} denotes the number of bins selected by the parameter ζ . Thus a motion difference vector $\Delta \mathbf{h}_i^\zeta$ for the i^{th} individual with its M neighbors may be represented as $\Delta \mathbf{h}_i^\zeta = [\Delta h_{i,1}^\zeta, \Delta h_{i,2}^\zeta, \dots, \Delta h_{i,M}^\zeta]$, where $j = 1, \dots, M$.

Algorithm 2: Frame Anomaly detection

Input: Video Sequence $\{frame^t\}, \beta$
Output: $frame^t \leftarrow \Omega, \bar{\Omega}$
1: *Initialization:* $M \leftarrow 5, \alpha \leftarrow 70, \zeta \leftarrow 0.9$
2: **for** $t = 1$ **to** T **do** * T is the total No. of frames */
3: Compute OF map $\{u(x, y), v(s, y)\}$
4: Compute Saliency map $\{S(x, y)\}$
5: Compute MOF map $\{u'(x, y), v'(x, y)\}$ using Eq. 9
6: **for** $i = 1$ **to** β **do**
7: Determine M nearest neighbor
8: Build motion descriptor $\{W_i, F_i\}_{k=1}^M$ from Eq. 11 & 12
9: Compute FA_{raw}^t over β observers
10: Obtain filtered EMD as FA^t using Eq. 14
11: **if** $FA^t > threshold$ **then**
12: $frame^t \leftarrow \Omega$
13: **else**
14: $frame^t \leftarrow \bar{\Omega}$
15: **end if**
16: **end for**
17: **end for**

3.2.3. Local descriptor construction

Several local motion descriptors are constructed from the above motion statistics for each of the β observers (described in Section 3.1.3) with its surrounding M neighbors based on $\Delta \mathbf{h}_i^\zeta$, where, $i = 1, \dots, \beta$. A local descriptor (see Fig. 5) is represented as $\{\mathbf{W}_i, \mathbf{F}_i\}_{k=1}^M$, where, $\mathbf{W}_i \in \mathbb{R}^{1 \times M}$ is the linking weight vector of the i^{th} individual with its M neighbors and $\mathbf{F}_i \in \mathbb{R}^{4 \times M}$ is their four-dimensional attributes of motion statistics. Each of these four attri-

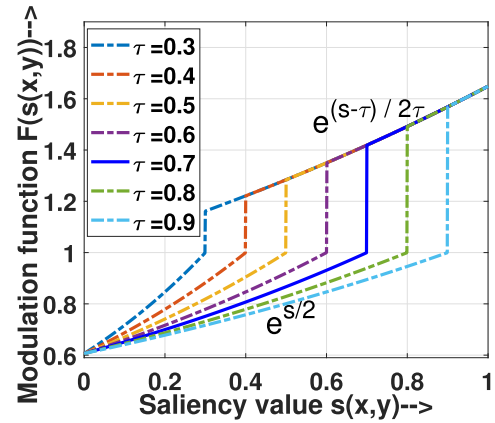


Fig. 4. A family of modulating function $\mathcal{F}(s)$ for different values of τ . Based on τ flow vectors are either enhanced or diminished.

butes are maximum, minimum, mean and variance of motion energy of flow vectors enclosed within ROI defined by their corresponding BBs. A linkage weight is constructed for each observer i to its k^{th} ($k \in M$) surrounding neighbor as:

$$W_{i,k} = \frac{(\Delta h_{i,k}^\zeta * D_{i,k}^{RCNN})^2}{\sum_{k=1}^M (\Delta h_{i,k}^\zeta * D_{i,k}^{RCNN})^2} \quad (12)$$

where, $D_{i,k}^{RCNN}$ is the Euclidean distance between semantic features of the i^{th} observer and its k^{th} neighbor and is extracted from the RCNN (please see Eq. 4). From the above equation it can be seen that the linkage weight is a normalized product of RCNN feature distance ($D_{i,k}^{RCNN}$) and motion difference vector ($\Delta \mathbf{h}_i^\zeta$) over M neighbors scaled over the quadratic function. Thus, the more distinct is the i^{th} observer from its k^{th} neighbor (based on both motion and semantic similarity), the larger will be their linkage weight.

3.3. Crowd anomaly analysis

The feature attributes of M neighbors along with their linkage forms a distinctive local descriptor with an observer. The descriptors of the corresponding observers are generated and compared across frames to measure magnitude of local disruption. An aggregate disruption of all the local descriptors contributes to the overall local change in a frame. If the magnitude of these local changes are beyond a certain limit then that frame is treated as anomalous. As EMD measure is better at evaluating dissimilarity between two multi-dimensional distributions, it is utilized to measure changes between local descriptors. The relative variation between local

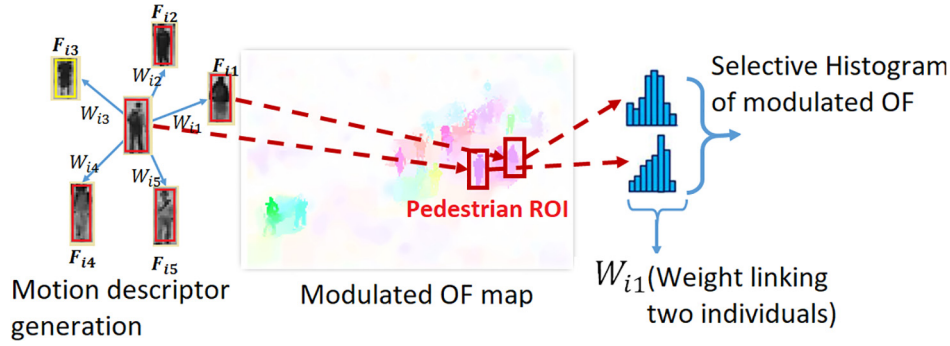


Fig. 5. A schematic diagram showing motion descriptor generation using SHMOF computed from MOF map within ROI of detected object within bounding boxes.

descriptors of the i^{th} observer at frame $(t-1)$ (i.e., $\{\mathbf{W}_i^{t-1}, \mathbf{F}_i^{t-1}\}_{k=1}^M$) and at frame t (i.e., $\{\mathbf{W}_i^t, \mathbf{F}_i^t\}_{k=1}^M$) can be expressed as:

$$FA_{raw}^t = \frac{1}{\beta} \sum_{i=1}^{\beta} EMD(\mathbf{W}_i^{t-1}, \mathbf{W}_i^t, \mathbf{F}_i^{t-1}, \mathbf{F}_i^t) \quad (13)$$

where, FA_{raw}^t , computed for the t^{th} frame, is termed as raw abnormality value and is often stochastic in nature. Sometimes, a high FA_{raw}^t value can result from erroneous detection and can also render a normal frame abnormal. Therefore, a post-filtering operation is performed to smooth out this effect using a weighted mean filter. Thus final abnormality measure is:

$$FA^t = \frac{1}{2n+1} \sum_{t'=(t-n)}^{t+n} wt * FA_{raw}^{t'} \quad (14)$$

where, wt is a partial weight factor given by

$$wt = \begin{cases} \frac{1}{n+1} & \text{if } t' = n \\ \frac{1}{2(n+1)} & \text{if } t' \neq n \end{cases} \quad (15)$$

The filter length is $(2n+1)$, where, n is an integer. We have used $n=3$ in this paper. A frame is deemed anomalous (Ω) if it yields a FA^t value which exceeds 0.5 and is treated normal ($\bar{\Omega}$) otherwise. The threshold value of 0.5 is set according to [45]. Moreover, the EMD response of observers that are above this threshold value are localized as anomalous regions (see Fig. 11 in this connection). Unlike other forms of anomalies in video scenes [1,48], crowd anomaly mostly encounters complex scene dynamics. Non-stationarity of anomaly in both spatial and temporal regions are one of the key concerns addressed in this work. To handle such problems, we generate reliable motion descriptors via continually destroying and creating new template pools over stable observers (Section 3.1.3). Moreover, as a stable object (ob-

server) shifts spatio-temporally over the frame, the descriptors are generated in corresponding regions. This is illustrated in Fig. 6, 7. Thus our model can automatically handles non-stationarity in crowd anomaly.

4. Experimental results

4.1. Dataset description, evaluation metric and hyper-parameter settings

Dataset Description We have experimented with four publicly available datasets, namely, UCSD, UMN, CUHK-Avenue and ShanghaiTech. UCSD dataset [56] is mostly adopted for crowd anomaly. Common anomalies include entering of undesirable objects like bikers, skaters and small carts into a pedestrian walkway. Other instances of anomalies include unusual variation of crowd density. Unlike UCSD, UMN dataset [45] only consists of persons and anomaly described in this dataset points to a crowd escape behavior. CUHK Avenue [60] is another dataset with abnormal events such as loitering and running, throwing objects. We have also evaluated our model over a recently proposed Shanghai Campus dataset [2]. It consists of many scenes with complex viewing angles and lightning conditions.

Evaluation Metric The two most common metric adopted in literature [5,60] for evaluating performance of anomaly detection are receiver operating characteristics (ROC) and Area under the (ROC) curve (AUC). ROC represents the variation of true positive rate (TPR) vs false positive rate (FPR) of frames detected as anomalous compared to ground truth. ROC is computed by gradually changing the threshold values over anomaly scores/responses. Equal error rate (EER) is another measure adopted in some works [15,45] to compare the detection performance. EER summarizes the ratio of misclassified frames at which $FPR = 1 - TRP$. Higher AUC and lower EER values indicate better performance. Quantitative results

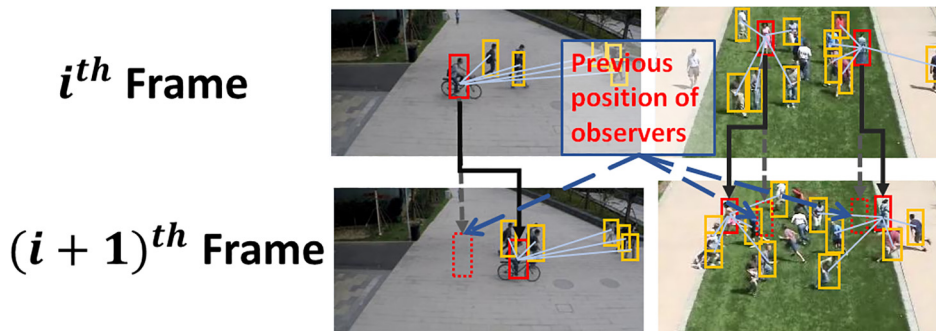


Fig. 6. The figure shows the non-stationarity of local descriptors where observers with their surrounding individuals are represented by red and yellow bounding boxes respectively. The dotted bounding boxes in frame 2 show the previous position of the observers.

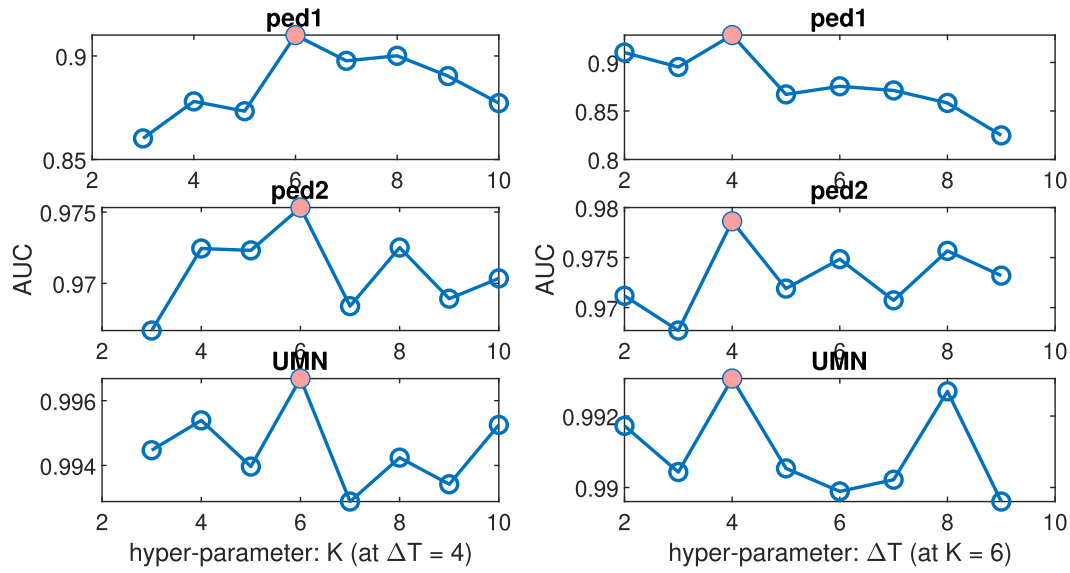


Fig. 7. Effect of different hyper-parameters on AUC over UCSD ped1, UCSD ped2 and UMN dataset.

are shown in Fig. 8 and Fig. 9. Qualitative comparisons on anomaly detection are shown in (Fig. 11). We have also computed IoU (Intersection over union) [62] to evaluate association accuracy of

our 3D-DCT of detected BB in current frame with the previous ones. For multiple detections we take the mean of all IoUs (mIoU)

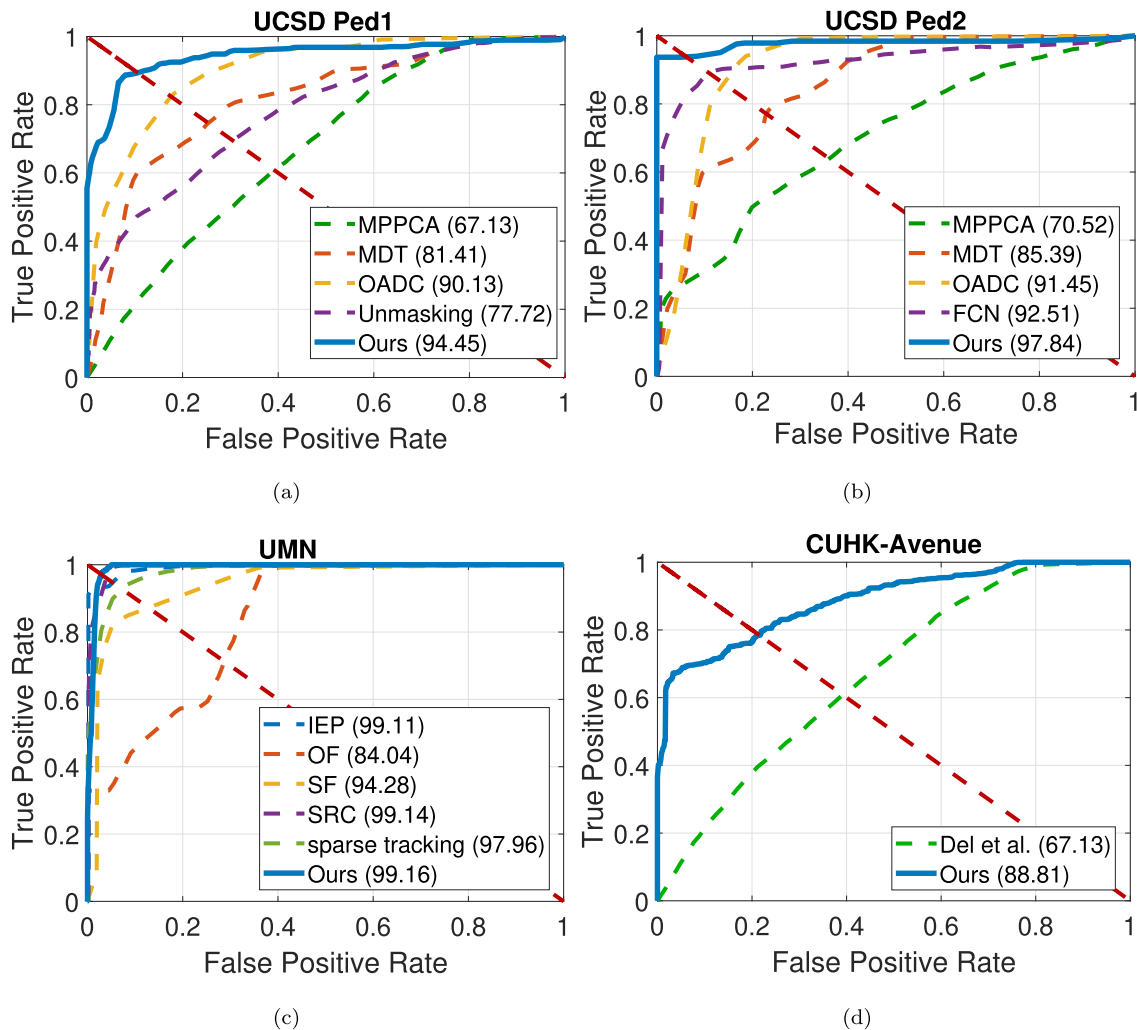


Fig. 8. Comparisons of Frame-level ROC over dataset UCSD ped1, UCSD ped2 dataset and UMN video sequences in figures (a), (b) and (c) respectively.

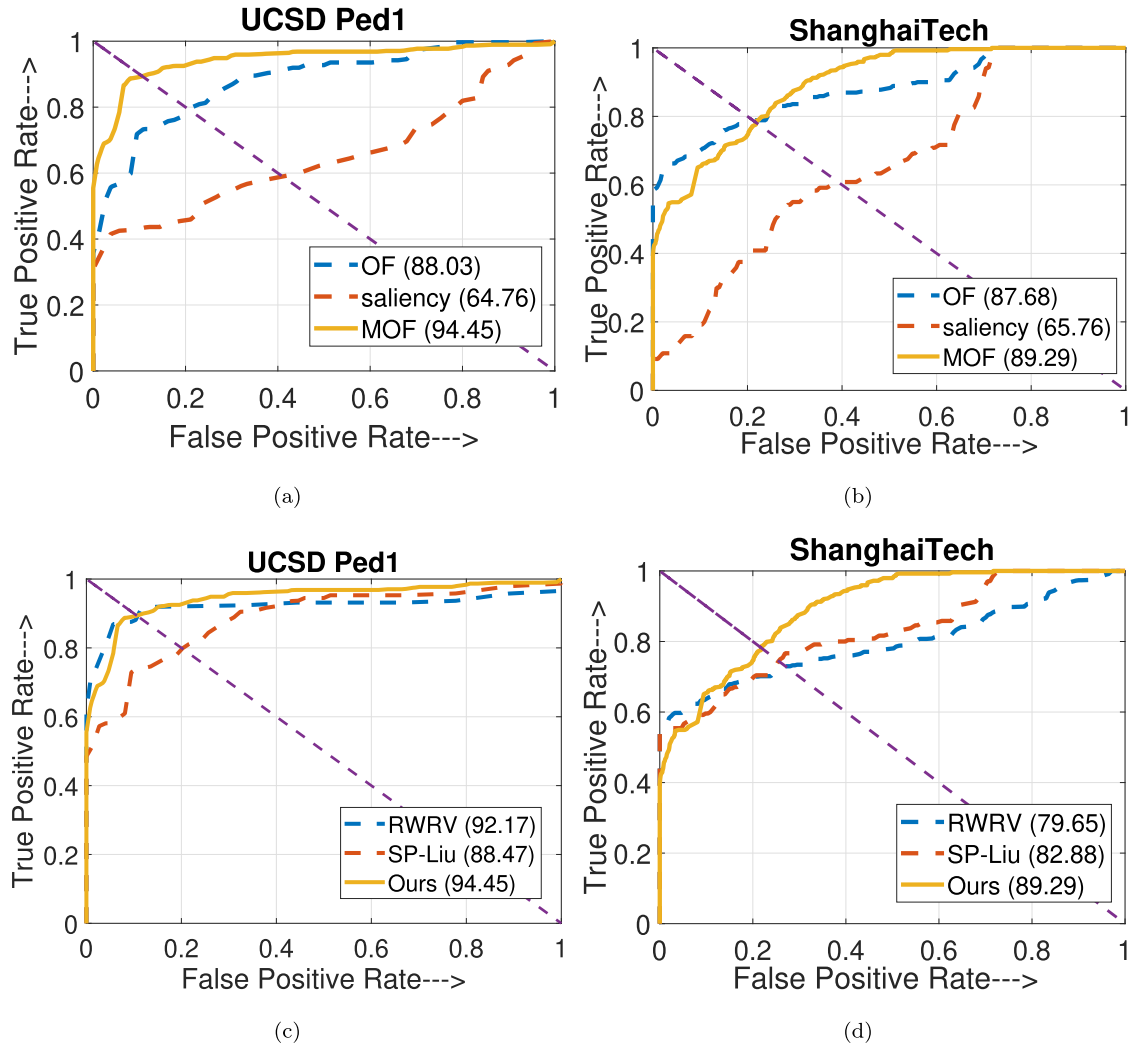


Fig. 9. Ablation study over UCSD Ped1 and ShanghaiTech dataset, where, figure (a) and (b) shows ROC curves using OF, MOF and saliency map. Figures (c) and (d) shows performance with other temporal saliency maps (RWRV [61] and SP-Liu [57]).

over all frames in a video. In general an $\text{IoU} > 0.5$ is considered to be a good association.

Hyper-parameter Settings There are four hyper-parameters for tuning model performance, i.e., ϕ_{thres} , ψ_{thres} , ΔT and K . While the first two parameters are related to proposal BB generation the rest are for maintaining reliability of 3D-DCT based association over video frames. In general, the aspect ratio of a pedestrian is set to ~ 0.41 ([63]). So, we have kept the aspect ratio of proposals within a range of $[0.33, 0.49]$. We set the values of ϕ_{thres} and ψ_{thres} to 0.5 and 0.7 based on experiments shown in Fig. 10, 11 and Table 5 respectively. This setting works for most of the video scenes. For the last two parameters, we analyzed the AUC performance as shown in Fig. 7. By analyzing the curves we set $K = 6$ and $\Delta T = 4$.

4.2. External comparisons on different datasets

Results on UCSD dataset Our results on UCSD Ped1 and UCSD Ped2 datasets are compared with several state-of-the-art methods like MPPCA [42], MPPCA + SFA [5], MDT [5], OADC-SA [45], Conv-AE [38], ConvLSTM-AE [32], Unmasking [26], Stacked RNN [2], Liu et al. [39], Sabokrou et al. [15] and Ionescu et al. [19], Liu et al. [39], Ullah et al. [17], Nawaratne et al. [20], Bansod et al. [18], Khan et al.

[34], Zhou et al. [35] and Singh et al. [36] (see Table 2 in this connection). Figs. 7 and 8(b) show comparison of ROCs of our model with different methods for Ped1 and Ped2 datasets respectively. AUC and EER values are reported in Table 2. Our method has achieved an overall AUC of 94.5% and 97.84% in UCSD Ped1 and Ped2 respectively which is highest among all the competing methods. The second best performing method in UCSD Ped1 is Singh et al. [36] (AUC = 93.2%) while in UCSD Ped2 is Ionescu et al. [19] (AUC = 97.8%). Our performance based on EER in Ped1 and Ped2 dataset are 10.6% (2nd best is OADC-SA = 18.1%) and 6.2% (2nd best is Zhou et al. = 10.3%) respectively which are also lower than other competing methods. Qualitative results on some sample video clips of Ped1 and Ped2 are shown in Fig. 11. Additionally, we have also performed some comparative study with temporal saliency maps of other methods, namely, SP-Liu [57] and RWRV [61] over UCSD Ped1 which have attained an AUC of 88.47% and 92.17% respectively. In our case modulation with temporal saliency map [58] have gained in AUC by 6.03% and 2.33% over SP-Liu and RWRV respectively (see Fig. 9(c)).

Results on UMN Dataset: Methods based on OF [9], SF [9], Interaction energy potentials (IEP) [30], SRC [4], tracking sparse components [31], OADC-SA [45], Ravanbakhsh et al. [14], Ionescu et al.

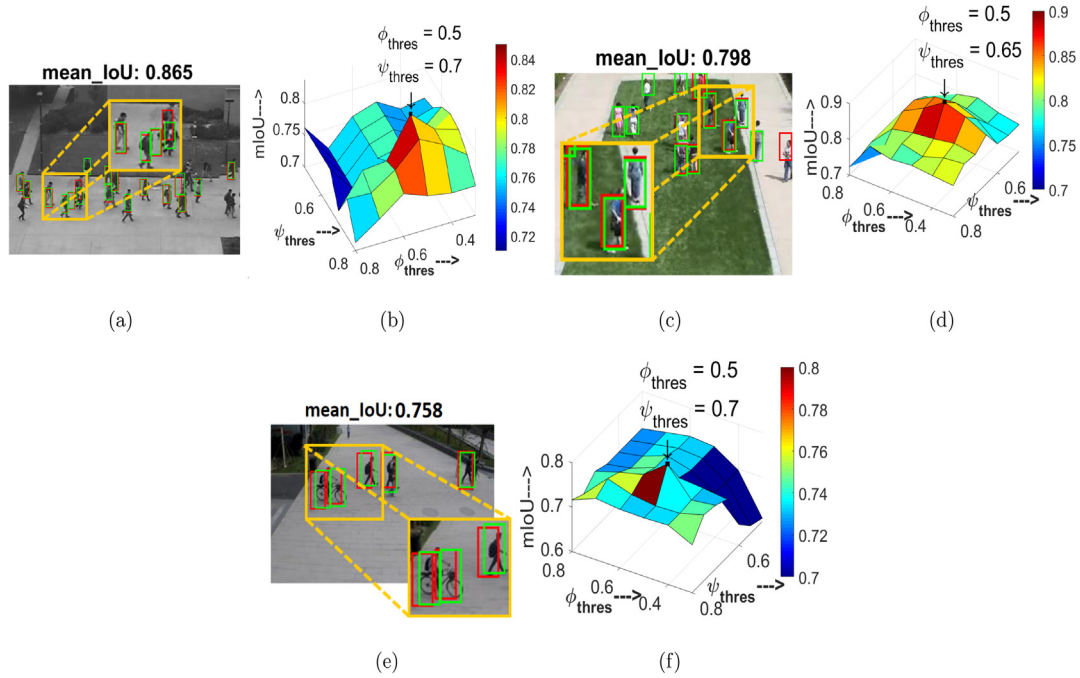


Fig. 10. Qualitative results showing performance of adaptive 3D-DCT based association based on mIoU for different ϕ_{thres} and ψ_{thres} values. Here, (a), (c), (e) are frames showing overlap of current detection with previous ones and (b), (d), (f) showing optimal parameters for UCSD, UMN and ShanghaiTech dataset respectively.

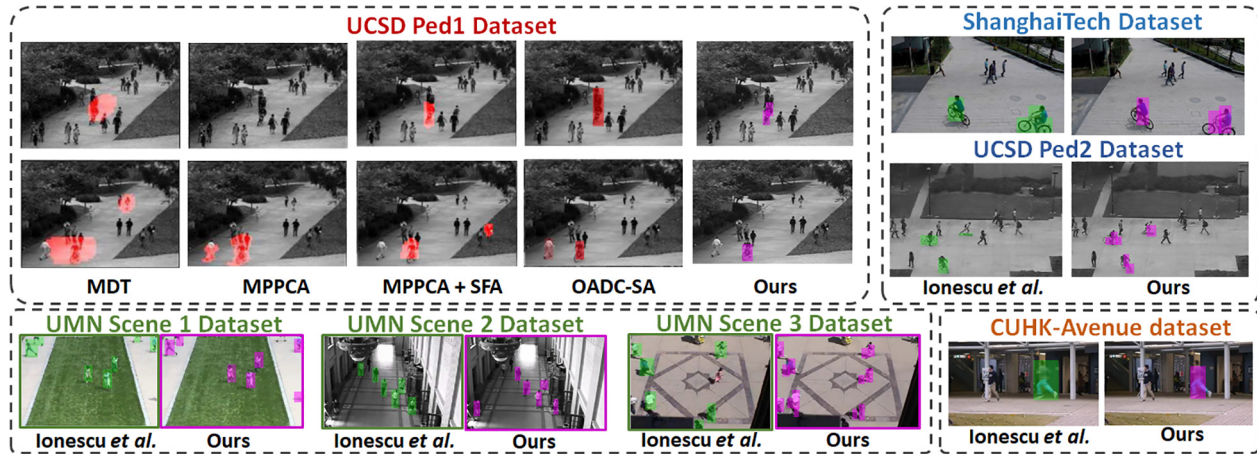


Fig. 11. Qualitative comparisons of anomaly detection over UCSD, UMN, CUHK-Avenue and ShanghaiTech datasets. We have compared with methods (for which qualitative results are available) like MDT [5], MPPCA [42], MPPCA + SFA [5], OADC-SA [45] for UCSD Ped1 dataset and Ionescu et al. [19] in UCSD Ped2, UMN, CUHK-Avenue and ShanghaiTech dataset. For most methods either there are missed anomalous regions or false anomalous detections where as our detections are more accurate.

[19], Bansod et al. [18], Khan et al. [34] and Hatirnaz et al. [41] are the competing methods on the UMN dataset. Fig. 8 shows comparison of ROC curves of these methods with that of ours. In Table 3, we separately report AUCs for 3 different UMN scenes, i.e., scene 1, scene 2 and scene 3 as 97.37%, 98.99%, 98.95% respectively and an overall AUC of 99.16%. Our overall AUC is only slightly less than the best method OADC-SA [45] (AUC = 99.67%) but is competitive otherwise. For association accuracy, we have attained the highest mean IoU (mIoU) value of 0.882.

Results on CUHK-Avenue dataset: Many recent methods, such as, Del et al. [25], Conv-AE [38], Unmasking [26], Stacked RNN [2], Liu et al. [39], Ionescu et al. [19], Nawaratne et al. [20], Zhou et al. [35] and Singh et al. [36] have reported their results on this relatively new dataset. In Fig. 8(d), we have shown ROC comparison with Del et al. [25] only as other methods have not provided ROC curves. In terms of AUC values (Table 2), we have performed exceedingly

well (with AUC = 88.81%) in comparison to Del et al. (AUC = 78.3%) and Conv-AE [38] (AUC = 80.8%) which are the most recognized methods in this dataset. Our method falls slightly below Ionescu et al. (AUC = 90.4%) and Singh et al. (AUC = 89.3%) by 1.59% and 0.49% respectively. A slight lag in performance in this dataset is due to sparse nature of dataset and difficulty in constructing local descriptor (see Section 3.2.3). Please note that we have attained such performance using a training-less model while most of the other methods are supervised.

Results on ShanghaiTech Campus Dataset: ShanghaiTech Campus dataset is another newly introduced relatively more complex dataset with much diverse lightning and viewing conditions. In this dataset, we have achieved best results with an AUC value of 89.29%. Here, we have outperformed methods like Conv-AE [38], Stacked RNN [2], Liu et al. [39] and Ionescu et al. [19] with corresponding AUCs as 60.9%, 68.0%, 72.8% and 84.9%.

Table 2

AUC (%) and EER (%) comparisons on UCSD Ped1, UCSD Ped2, CUHK-Avenue and ShanghaiTech datasets.

Datasets	UCSD Ped1		UCSD Ped2		CUHK Avenue	ShanghaiTech
Methods ↓ Criteria →	AUC	EER	AUC	EER	AUC	AUC
MPPCA [42]	59.0	32.0	69.3	36.6	–	–
MPPCA + SFA [5]	66.8	32.0	61.3	36.0	–	–
MDT [5]	81.8	25.0	82.9	25.0	–	–
OADC-SA [45]	91.0	18.1	92.5	14.1	–	–
Del et al. [25]	–	–	–	–	78.3	–
Conv-AE [38]	75.0	–	85.0	–	80.8	60.9
ConvLSTM-AE [32]	75.5	–	81.1	–	77.0	–
Unmasking [26]	68.4	31.4	82.2	–	80.6	–
Stacked RNN [2]	–	–	92.2	–	81.7	68.0
Liu et al. [39]	83.1	–	95.4	–	85.1	72.8
Sabokrou et al. [15]	–	–	92.5	11.0	–	–
Ullah et al. [17]	–	16.5	–	17	–	–
Nawaratne et al. [20]	75.2	29.9	91.1	8.9	76.8	–
Bansod et al. [18]	82.31	21.43	94.16	13.25	–	–
Khan et al. [34]	81.08	23.7	93.75	9.8	–	–
Zhou et al. [35]	83.5	25.2	94.9	10.3	86.1	–
Ionescu et al. [19]	–	–	97.8	–	90.4	84.9
Singh et al. [36]	93.2	–	92.1	–	89.3	–
Our Proposed	94.5	10.6	97.84	6.2	88.81	89.29

Table 3

AUC (%) comparisons on different scenes of UMN dataset.

Methods	OF [9]	SF [9]	IEP [30]	SRC [4]	Tracking sparse	OADC-	Ravanbakhsh	Ionescu	Bansod	Khan	Hatirnaz	Ours
					comp. [31]	SA [45]	et al. [14]	et al. [19]	et al. [18]	et al. [34]	et al. [41]	
Scene1	–	–	–	99.5	99.97	–	–	–	93.96	99.58	99.33	99.37
Scene2	–	–	–	97.5	93.1	–	–	–	98.26	97.49	94.72	98.99
Scene3	–	–	–	96.4	84.3	–	–	–	92.04	98.5	97.92	98.95
Overall	84.0	96.0	98.9	–	–	99.67	99.0	99.6	–	98.19	97.32	99.16

Table 4

Showing the variance of normalized likelihood value and AUC over fixed and adaptive 3D-DCT model.

Dataset	Variance of likelihood ($\rho(\mathcal{L})$)				AUC (%)	
	Frame 5		Frame 10		Fixed	Adaptive
	Fixed 3D-DCT	Adaptive 3D-DCT	Fixed 3D-DCT	Adaptive 3D-DCT	3D-DCT (ξ) [45]	3D-DCT (ξ) (ours)
UCSD ped1	0.00112	0.00407	0.00136	0.00143	85.11	94.57
UCSD ped2	0.00193	0.00498	0.00193	0.0035	89.64	97.84
UMN	0.00157	0.00484	0.00208	0.00396	90.27	99.16
CUHK-Avenue	0.00128	0.00479	0.00227	0.00538	79.98	88.81
ShanghaiTech	0.00098	0.00405	0.00159	0.00319	83.45	89.29

Table 5Results showing mIoU for different values of ϕ_{thres} and ψ_{thres} for UCSD, UMN and ShanghaiTech datasets.

Datasets	UCSD Ped2 (mIoU)								UMN (mIoU)								ShanghaiTech (mIoU)							
$\phi_{thres} \downarrow \psi_{thres} \rightarrow$	0.5	0.55	0.6	0.65	0.7	0.75	0.8	0.5	0.55	0.6	0.65	0.7	0.75	0.8	0.5	0.55	0.6	0.65	0.7	0.75	0.8			
0.3	0.753	0.759	0.783	0.794	0.791	0.773	0.746	0.775	0.775	0.782	0.837	0.812	0.817	0.786	0.608	0.618	0.647	0.731	0.744	0.749	0.709			
0.4	0.747	0.749	0.749	0.761	0.803	0.800	0.746	0.785	0.785	0.828	0.864	0.820	0.810	0.791	0.697	0.697	0.700	0.737	0.737	0.734	0.734			
0.5	0.769	0.769	0.769	0.765	0.838	0.820	0.753	0.804	0.804	0.850	0.882	0.810	0.791	0.796	0.732	0.732	0.732	0.742	0.799	0.732	0.732			
0.6	0.757	0.757	0.757	0.757	0.775	0.783	0.767	0.791	0.791	0.844	0.843	0.804	0.804	0.790	0.736	0.736	0.746	0.746	0.757	0.739	0.732			
0.7	0.719	0.719	0.719	0.719	0.771	0.758	0.739	0.752	0.792	0.812	0.812	0.792	0.752	0.804	0.726	0.726	0.726	0.726	0.744	0.756	0.738			
0.8	0.763	0.763	0.763	0.763	0.719	0.759	0.713	0.725	0.725	0.725	0.732	0.732	0.723	0.723	0.716	0.716	0.716	0.716	0.727	0.702	0.716			

The performance of our model is comparable to method [19] in UCSD Ped2 and slightly lagging in CUHK-Avenue dataset (see Table 2 in this connection). In spite of this performance lag our model possess certain other advantages with respect to [19]. Firstly, the model described in [19] requires two stages of training. They trained an object-centric convolutional auto-encoder (CAE) following a SVM classifier for separating normal and abnormal

frames. For such training both object-level annotation and frame-label annotations are required whereas our model, being a training-less one, has no such requirements. Secondly, their model requires explicit training for different datasets/scenes which restricts their application to any practical scenarios where availability of training data cannot be ensured. So, we have an implicit advantage as a training-less model. Furthermore, they have

achieved such training utilizing high computing resources (like high-end GPUs) [19] whereas our model can simply run on a CPU which can be further accelerated using any GPU.

The major weakness of the proposed approach is that the model performance can suffer from less accurate construction of the local descriptor (see Section 3.2.3) with increasing sparsity of crowded scenes. CUHK-Avenue is one such scenario where [19] yields better performance than ours (this is shown in Table 2). Additionally, our model performance is susceptible to motion artifacts that may arise due to inaccurate foreground mask generation by GMM which is less likely in [19] due to more accurate object detection.

4.3. Ablation studies

4.3.1. Influence of modulation

We have evaluated our model performance using optical flow (OF), saliency [58] and our proposed Saliency modulated OF (MOF) map exclusively to show the effectiveness of modulation. Performance is analyzed based on ROCs and AUCs computed over UCSD and ShanghaiTech datasets. See Figs. 8(d), and 9(b) in this connection. As shown in these figures, the influence of modulation is much more pronounced with an overall AUC of 94.45% when using MOF map as compared to only OF (AUC = 88.03%) and only saliency map (AUC = 64.76%) on UCSD Ped1 dataset. Also, on ShanghaiTech dataset, our approach with MOF has achieved an AUC of 89.29% as compared to with only OF (AUC = 87.68%) and with only saliency (AUC = 65.76%) map.

4.3.2. Impact of adaptive 3D-DCT based association

In Table 4, we have shown impact of adaptive ξ over fixed ξ (fixed at 0.1 according to OADC-SA [45]) on the overall performance in terms of AUCs and variance of likelihood measure, i.e., $var(\rho(\mathcal{L}))$ (see in Eq. 2). From the table, it can be seen that UCSD Ped1 has achieved highest gain attaining an AUC of 94.57% (using adaptive 3D-DCT), compared to that of 85.11% with fixed 3D-DCT. The lowest but a significant gain in AUC of 89.29% has been achieved using adaptive 3D-DCT while AUC of 83.45% is achieved using fixed 3D-DCT. AUC gains on other datasets are within a range of 5.84–9.46% on all other datasets. Furthermore, an increased likelihood variance ($var(\rho(\mathcal{L}))$) in case of adaptive 3D-DCT can be observed in Table 4. We have computed $var(\rho(\mathcal{L}))$ for the 5th and the 10th frame for a randomly selected video sequence from each dataset in a completely unbiased manner. It can be clearly seen that the variance of the likelihood values is significantly higher for the estimated ξ in all cases.

4.3.3. Impact of RCNN based semantic feature fusion

We have also performed an ablation study for RCNN based semantic features (see Fig. 12). As shown in this figure, the improvement in AUC is mostly prominent in more complex scenarios of ShanghaiTech (improved from 76.8% to 88.81%), CHUK-Avenue dataset (from 80.99% to 81.29%) and UCSD Ped1 dataset (from 88.2% to 94.5%); whereas, in scenes of UCSD Ped2 (from 94.2% to 97.4%) and UMN (from 98.2% to 99.16%), minor but still improved AUC values are noticed when RCNN features are incorporated in the model. Such results clearly indicate that the impact of semantic feature mining and utilization on the performance in a training-less system.

4.3.4. Comparison of different saliency maps

Comparisons of the saliency model used in this work ([58]) with two other temporal saliency methods, namely, SP-Liu [57] and RWRV [61] have been carried out over the UCSD Ped1 and ShanghaiTech datasets (see Fig. 9(c), (d)). SP-Liu and RWRV have achieved an AUC of 88.47% and 92.17% in UCSD Ped1 dataset whereas we have attained 94.45%. In ShanghaiTech dataset, we have achieved 89.29% exceeding SP-Liu and RWRV by 6.41% and 9.64% respectively. The above results clearly indicate the effectiveness of our temporal saliency map.

4.4. Run-time analysis

Computation time plays a significant role in video surveillance but only a few methods have reported it. We have conducted all of our experiments on MATLAB 2018a platform installed on a PC with 64-bit windows OS with the processor speed of 3 GHz and 8 GB RAM. As reported in [45], the average run-time per frame for MDT [5] is 25 s on a standard platform of 3 GHz processor (which is close to ours) for UCSD dataset. For SRC [4] the average time spent on each frame is 0.8 s for UMN dataset. The average time of OADC-SA is 0.6 s on UCSD and UMN dataset. Unmasking [26] and Stacked RNN [2] have achieved a computation time of 0.5 s and 0.2 s respectively but Stacked RNN required additional time for training the neural network. Our method, without involving any training phase, required on average 2.87 s and 4.72 s to process a frame on UCSD and UMN dataset respectively. For CUHK-Avenue and ShanghaiTech it took approximately 2.25 s and 2.58 s to process a frame. Please note that such computation times have been achieved without any code optimization.

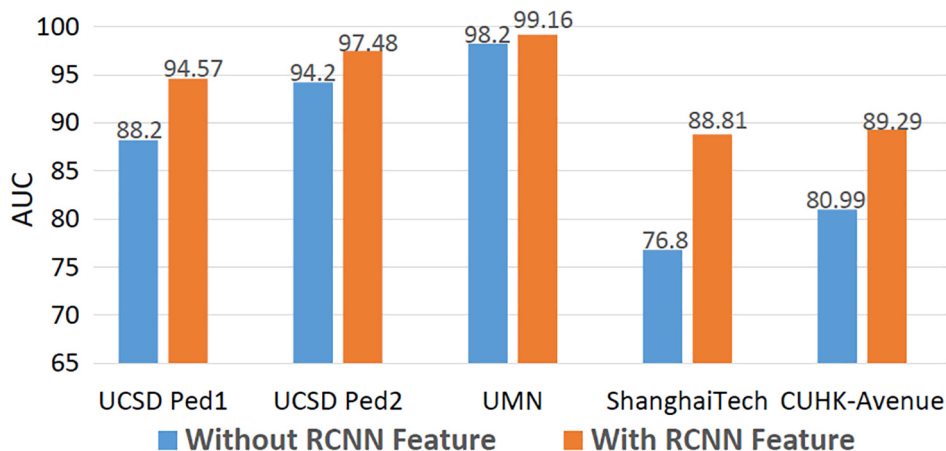


Fig. 12. Comparison of our model performance (in terms of AUC values) with and without (D_{ik}^{RCNN}) feature.

5. Conclusion

In this paper, we proposed an adaptive training-less method for anomaly detection. Our model determines anomaly without any prior training while learning certain model parameters on-the-fly. This approach is quite useful for surveillance problems where any prior assumption of the anomalous scenes could be misleading. Our approach has achieved competitive results compared to current state-of-the-art methods on four publicly available UCSD, UMN, CUHK-Avenue, ShanghaiTech datasets. In future, we plan to make our solution robust to motion artifacts. Another direction of future research will be to improve the model for crowded scenes with high sparsity.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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